# Aleksei Volodin

Senior Android Engineer · AI Native Engineer

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## SUMMARY

Senior Android engineer and AI Native Engineer with 20+ years in software and 12+ on Android. Key contributor to two high-traffic Chilean banking apps with more than a million installs each. Builds on-device AI in the open with Knotwork, a local-first agent for Android, developed spec-first with coding agents under human review. Has shipped companion apps for BLE and Wi-Fi hardware, keeps crash-free sessions above 99% across releases, and has worked in distributed teams for 15 years.

## EXPERIENCE

### Senior Android Developer, Muruna — Jun 2022 – now

Remote, Santiago, Chile. FinTech, two retail banks

- New features across retail-banking Android modules, with Clean Architecture, MVVM and Hilt or Dagger.
- API migrations and networking refactors on Retrofit and OkHttp, keeping backward compatibility.
- Resolved defects from internal and external stakeholders through structured Kotlin and Java debugging.
- Maintained unit and UI tests with JUnit and Espresso, and static checks with Detekt and Ktlint.

### Senior Android Developer, MOBDEV — Feb 2021 – Jun 2022

Remote, Santiago, Chile. FinTech, a retail bank

- Android features in Kotlin with Jetpack Navigation, WorkManager, Retrofit and OkHttp.
- Features and refactors within an established MVI architecture.
- Worked with Product and UX on key flows: sign-in, onboarding and payments.
- Diagnosed issues with logs, crash reports and profilers, then shipped targeted fixes across modules.

### Senior Android Developer (contract), Swing AI & LifeJourney — Jul 2017 – Mar 2020

Remote. One team across both contracts

- Built and evolved Android applications with a focus on reliable delivery.
- Worked with design, backend and QA, with close attention to UX and performance.
- Created and maintained unit and UI tests around critical flows.

### Lead Android Developer, Knotable — Sep 2014 – Jan 2017

Remote, for a New York company

- Designed and built two Android applications and led a team of three engineers.
- Improved reliability and performance through targeted refactors and systematic debugging.

### Senior Android Developer, Freelance — Sep 2011 – now

Remote, Santiago, Chile

- More than 20 Android apps: streaming media, a layered image editor, live video, project management, chats, readers, enterprise clients, inventory tools.
- End-to-end delivery from requirements to release.
- Research and integration of on-device AI and agentic development workflows.

### Software analyst and engineer, 1C · C++ · SQL — 2003 – 2012

Earlier career, condensed

## SELECTED PROJECTS

### Knotwork

The model runs on the phone through LiteRT-LM, and every request goes through a pipeline you can read and change. By default a risky action waits for your confirmation.

Stars 51 Commits 2076 Merged pull requests 613 Tagged releases 21 Latest release 0.14.0 License Apache-2.0

GitHub Google Play F-Droid

### Colored Seeds

Match pairs of colored seeds, clear rows and empty the board. A long-running game, rewritten and relaunched as 3.0 in 2026.

Downloads 50K+ Rating 4.0 Ratings 551 Last update Oct 31, 2026

Google Play Site

## AI NATIVE ENGINEERING

Coding agents work across the whole SDLC. The engineer owns specification, architecture and release decisions.

**Spec-Driven Development (SDD):** Each task starts with a written specification: affected classes, interfaces, tests and files. Implementation begins only after it is approved.

**AI-Driven Development (AIDD):** Agents carry research, implementation, tests, review and refactoring through pull requests. Official documentation is checked before any library or API is used.

**Human-in-the-loop gates:** Stage gates at plan approval, design handoff, pull-request review and release. Irreversible actions require explicit human approval.

**Context engineering:** An agent manifest as the source of truth, path-scoped rules, a generated code map, and long-term memory retrieved at session start.

**Multi-agent orchestration:** Subagents for parallel research and independent code review, reusable commands and skills, tools connected over MCP.

**Guardrails as code:** Pre-execution hooks on shell, git and pull-request operations. An LLM gate rejects commits and pull requests whose claims name no evidence.

**Agentic code review:** The agent re-reads the real diff as a reviewer against a checklist. Each defect gets a regression test, and the pass repeats until it finds nothing new. A human review follows.

**CI quality gates:** Detekt, Ktlint, Android Lint, a 75% Kover coverage floor, Konsist architecture tests, Roborazzi screenshot regression and instrumented tests.

**ADR and docs as code:** Architecture Decision Records for every significant decision. Documentation changes ship in the same pull request, with link checks in CI.

**Release engineering and supply chain:** Phase integration branches keep main releasable. Dependency verification with PGP and SHA-256, CI actions pinned to commit SHAs, secret scanning.

## SKILLS

**Android:** Kotlin, Java, Android SDK, Jetpack Compose, Architecture Components, Coroutines, Flow, Retrofit, OkHttp, WorkManager, Room, Navigation, Hilt, Dagger, Koin, RxJava, WebSocket, performance optimisation

**On-device AI:** LiteRT, LiteRT-LM, Koog, AppFunctions Jetpack, local LLM integration, privacy-first inference

**AI Native Engineering:** Spec-Driven Development, AI-Driven Development, agentic SDLC, context engineering, multi-agent orchestration, human-in-the-loop gates, guardrails as code, MCP, prompt engineering; Claude Code, GitHub Copilot, Gemini CLI

**Testing and QA:** JUnit, Espresso, UI Automator, Mockito, MockK, Detekt, Ktlint, CI test pipelines

**IoT / BLE:** GATT, connection reliability, background sync, OTA; UV sterilisers, smart locks, e-scooters

**Services and data:** REST, JSON, XML, Firebase, AWS, SQL, SQLite, Realm, MongoDB

**CI/CD and tools:** Gradle, GitHub Actions, Bitrise, Jenkins, Android Studio, IntelliJ IDEA

## EDUCATION AND LANGUAGES

**Education:** BS in Computer Science, Serpukhov Technical College, 2000–2003

**Certification:** GitHub Copilot, Microsoft, October 2026, valid until October 2028; Security Skilled, 2025

**Languages:** English, fluent. Spanish, fluent. Russian, native.